

Curriculum Vitae of **Nian-Feng Tzeng**

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PROFESSIONAL EXPERIENCE

Eminent Scholar Endowed Chair (August 2017 – Present), School of Computing and Informatics, University of Louisiana at Lafayette.

Associate Director for Academic Affairs, School of Computing and Informatics, UL Lafayette (August 2016 – May 2022).

Lockheed Martin Corporation / BoRSF Professor (August 2004 – July 2017), Center for Advanced Computer Studies, UL Lafayette.

Professor of Computer Engineering (August 1997 – July 2004), Center for Advanced Computer Studies.

Visiting Professor (August – December 2000), Sabbatical with Department of Computer Science, Rice University, Houston, Texas.

Senior Faculty Fellow (June – August 1997), Naval Surface Warfare Center, U.S. Navy-ASEE Summer Faculty Research Program.

Associate Professor of Computer Engineering (1992 – 1997), Center for Advanced Computer Studies.

Assistant Professor of Computer Engineering (1987 – 1992), Center for Advanced Computer Studies.

Member of Technical Staff (1986 – 1987), AT&T Bell Labs, Columbus, Ohio.

EDUCATION

Ph.D., University of Illinois at Urbana-Champaign, 1986, Computer Science.

M.S., National Taiwan University, 1980, Electrical and Computer Engineering.

B.S., National Chiao Tung University, Taiwan, 1978, Computer Science.

RESEARCH AREAS

- High-Performance Computer Architecture and Systems
- Computer Memory Systems
- Computer Communications and Networks
- Resilient Computing and Networked Systems

RESEARCH GRANTS AND CONTRACTS – Summary

- Total Amount: \$13.2+ Million (institutional matching excluded)
- Sponsors: National Science Foundation, U.S. Department of Energy, Louisiana Board of Regents (State of Louisiana), U.S. Army Research Office.

List of Research Grants and Contracts

27. National Science Foundation, Office of Integrative Activities (OIA), Established Program to Stimulate Competitive Research (EPSCoR), “RII Track-2 FEC: Precise Regional Forecasting via Intelligent and Rapid Harnessing of National Scale Hydrometeorological Big Data,” \$5,000,000, September 2020 through August 2025 (Lead PI & PD, jointly with 4 Co-PIs and 9 Other Investigators from Louisiana, Alabama, and Kentucky).
26. National Science Foundation, Division of Computer and Network Systems, “CSR: Small: Collaborative Research: Comprehensive Algorithmic Resilience (CAR) for Big Data Analytics,” \$265,999, September 2015 through August 2019.
25. National Science Foundation, Division of Computer and Communication Foundations, “SHF: Small: Cooperative Memory Expansion (COMEX) for Networked Computing Systems via Remote Direct Memory Access,” \$476,000, July 2014 through June 2018.
24. National Science Foundation, Division of Computing and Communication Foundations, “SHF: Small: Reliability Enhancement via Adaptive Checkpointing in Wireless Grids,” \$420,000, August 2009 through July 2014.
23. National Science Foundation, Division of Computer and Network Systems, “NEDG: Featherlight Information Network with Delay-Endurable RFID Support,” \$366,000, September 2008 through August 2013 (jointly with H. Wu).
22. BORSF, Board of Regents, State of Louisiana, “Acquisition of a Mobile Sensor System for Prototyping and Empirical Evaluation of Delay/Fault-Tolerant Mobile Sensor Networks (DFT-MSN’s),” \$65,000, June 2009 through June 2011 (jointly with H. Wu).
21. U.S. Department of Energy, Office of Science, “Ubiquitous Computing and Monitoring System (UCoMS) for Discovery and Management of Energy Resources – Phase II,” \$950,000, August 2007 through April 2012 (as lead principal investigator, together with other 12 co-principal investigators from UL, Lafayette, LSU, and Southern University, Baton Rouge).
20. Board of Regents, State of Louisiana, “Ubiquitous Computing and Monitoring System (UCoMS) for Discovery and Management of Energy Resources – Phase II,” \$1,200,000, August 2007 through April 2012 (State’s matching support for the above DoE funded project).
19. BORSF, Board of Regents, State of Louisiana, “Approaches for Improving Wireless Data Communications under High Congestion,” \$150,000, June 2005 through June 2008 (jointly with H. Wu).
18. U.S. Department of Energy, Office of Science, “Ubiquitous Computing and Monitoring System (UCoMS) for Discovery and Management of Energy Resources,” \$1,200,000, August 2004 through August 2007 (as lead principal investigator, together with other 9 co-principal investigators from UL, Lafayette, LSU, and Southern University, Baton Rouge).

17. Board of Regents, State of Louisiana, "Ubiquitous Computing and Monitoring System (UCoMS) for Discovery and Management of Energy Resources," \$1,200,000, August 2004 through August 2007 (State's matching support for the above DoE funded project).
16. National Science Foundation, Division of Computer-Communications Research, "Architectural Support for Scalable High-Speed Routers," \$300,000, September 2001 through August 2006.
15. National Science Foundation, Division of Computer-Communications Research, "Reconfiguration and Performance Issues in Software Distributed Shared Memory Systems," \$212,065, September 1998 through August 2002.
14. National Science Foundation, Major Research Instrumentation Program, "Acquisition of Networked Heterogeneous Computer Systems," \$102,400, September 1998 through August 2002.
13. BORSF, Board of Regents, State of Louisiana, "Instrumentation for Research on Communications and Networks," \$50,000, June 2000 through June 2001.
12. U.S. Army Research Office, Electronics Division, "High-Performance Wireless Information Systems," \$229,504, May 1998 through April 2001 (jointly with G.-L. Feng).
11. BORSF, Board of Regents, State of Louisiana, "High-Performance Wireless Information Systems," \$97,292, October 1998 through April 2001.
10. BORSF, Board of Regents, State of Louisiana, "Acquisition of Networked Computer Systems," \$60,300, June 1998 through June 1999.
9. LEQSF, Board of Regents, State of Louisiana, "Interoperable Underground Wavelet Transients Munitions Classification," \$150,000, June 1996 through June 1999 (jointly with H. Szu and M. A. Bayoumi).
8. National Science Foundation, Computer and Computation Research, "Investigation into Faulty and Incomplete Message-Passing Parallel Computers," \$60,030, August 1993 through July 1996.
7. LEQSF, Board of Regents, State of Louisiana, "The Design and Development of Next Generation Geographic Information Systems," \$63,000, June 1994 through June 1996 (jointly with N. Pissinou).
6. LaSER, NSF and Board of Regents, State of Louisiana, "Developing an Architectural Framework for Distributed GIS," \$55,000, June 1994 through June 1995 (jointly with N. Pissinou and R. Logan).
5. LEQSF, Board of Regents, State of Louisiana, "Parallelization for Finite-Element Computation," \$65,050, June 1993 through June 1995 (jointly with H.-C. Chen).
4. National Science Foundation, Microelectronic Information Processing Systems, "Improving the Communication Performance of Multiprocessors," \$98,015, May 1992 through April 1995.
3. LEQSF, Board of Regents, State of Louisiana, "Communication Performance Improvement in Multiprocessors under Nonuniform Traffic," \$55,597, June 1992 through June 1994.
2. LEQSF, Board of Regents, State of Louisiana, "Prototyping of Parallel Algorithms for Engineering Applications," \$150,000, June 1992 through June 1993 (jointly with M. A. Bayoumi and M. Montenyohl).
1. National Science Foundation, Microelectronic Information Processing Systems, "Yield and Reliability Considerations of VLSI/WSI Computing Systems," \$169,206, September 1988 through August 1990 (jointly with L.N. Bhuyan and T.R. Rao).

PUBLICATIONS

Refereed Journal Papers

The following publications appeared in leading archival journals after fully peer-reviewed.

58. A. Sultana, N. Pakka, F. Xu, X. Yuan, L. Chen, and N.-F. Tzeng, "Resource Heterogeneity-Aware and Utilization-Enhanced Scheduling for Deep Learning Clusters," *IEEE Transactions on Computers*, vol. 75, No. 6, pp. 2169 – 2184, June 2026.
57. Y. Zhang, B. Turney, P. Sigdel, X. Yuan, E. Rappin, A. Lago, S. Kimball, L. Chen, P. Darby, L. Peng, S. Aygun, Y. Tu, M. H. Najafi, and N.-F. Tzeng, "Regional Weather Variable Predictions by Machine Learning with Near-Surface Observational and Atmospheric Numerical Data," *IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS)*, vol. 63, Article No. 4701621, February 2025.
56. J. Lian, X. Yuan, J. Lou, L. Chen, H. Wang, and N.-F. Tzeng, "Room-Scale Location Trace Tracking via Continuous Acoustic Waves," *ACM Transactions on Sensor Networks*, vol. 20, no. 3, pp. 61:1 – 61:23, April 2024.
55. F. Qi, K. N. Smith, T. Lecompte, N.-F. Tzeng, X. Yuan, F. T. Chong, and L. Peng, "Quantum Vulnerability Analysis to Guide Robust Quantum Computing System Design," *IEEE Transactions on Quantum Engineering*, vol. 5, January 2024.
54. J. Lou, X. Yuan, P. Sigdel, X. Qin, S. Kompella, and N.-F. Tzeng, "Age of Information Optimization in Multi-Channel based Multi-hop Wireless Networks," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 5719-5732, October 2023.
53. T. Lecompte, F. Qi, X. Yuan, N.-F. Tzeng, M. H. Najafi, and L. Peng, "Machine Learning-Based Qubit Allocation for Error Reduction in Quantum Circuits," *IEEE Transactions on Quantum Engineering*, vol. 4, Paper No. 3101414, September 2023.
52. T. LeCompte, L. Peng, X. Yuan, and N.-F. Tzeng, "Protecting Synchronization Mechanisms of Parallel Big Data Kernels via Logging," *IEEE Transactions on Computers*, vol. 71, no. 9, pp. 2156 - 2162, September 2022,
51. J. Lian, X. Yuan, M. Li, and N.-F. Tzeng, "Fall Detection via Inaudible Acoustic Sensing," *ACM Journal Series, Proceedings of ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 3, pp. 114:1 - 114:21, September 2021.
50. J. Lou, X. Yuan, S. Kompella, and N.-F. Tzeng, "Boosting or Hindering: AoI and Throughput Interrelation in Routing-aware Multi-hop Wireless Networks," *IEEE/ACM Transactions on Networking*, vol. 29, no. 2, pp. 1008-1021, June 2021.
49. P. Sigdel, X. Yuan, and N.-F. Tzeng, "Realizing Best Checkpointing Control in Computing Systems," *IEEE Transactions on Parallel and Distributed Systems*, vol. 32, no. 2, pp. 315-329, February 2021.
48. P. Srinuan, X. Yuan, and N.-F. Tzeng, "Cooperative Memory Expansion via OS Support for Networked Computing Systems," *IEEE Transactions on Parallel and Distributed Systems*, vol. 31, no. 11, pp. 2650-2667, November 2020.
47. X. Xiang, P. Sigdel, and N.-F. Tzeng, "Bufferless Network-on-Chips with Bridged Multiple Subnetworks for Deflection Reduction and Energy Savings," *IEEE Transactions on Computers*, vol. 69, no. 4, pp. 577-590, April 2020.

46. P. Sigdel and N.-F. Tzeng, "Coalescing and Deduplicating Incremental Checkpoint Files for Restore-Express Multi-Level Checkpointing," *IEEE Transactions on Parallel and Distributed Systems*, vol. 29, no. 12, pp. 2713-2727, December 2018.
45. W. Shu and N.-F. Tzeng, "NUDA: Non-Uniform Directory Architecture for Scalable Chip Multiprocessors," *IEEE Transactions on Computers*, vol. 67, no. 5, pp. 740-747, May 2018.
44. X. Li, M. Amini, M. Bayoumi, N.-F. Tzeng, and R. Buyya, "Cost-Efficient and Robust On-Demand Video Transcoding Using Heterogeneous Cloud Services," *IEEE Transactions on Parallel and Distributed Systems*, vol. 29, pp. 556-571, March 2018.
43. W. Shu and N.-F. Tzeng, "Compressed Sharer Tracking and Relinquishment Coherence for Superior Directory Efficiency of Chip Multiprocessors," *IEEE Transactions on Computers*, vol. 66, no. 11, pp. 1975-1981, November 2017.
42. I. Jangjaimon and N.-F. Tzeng, "Effective Cost Reduction for Elastic Clouds under Spot Instance Pricing through Adaptive Checkpointing," *IEEE Transactions on Computers*, vol. 64, no. 2, pp. 396-409, February 2015.
41. J. Maneesilp, C. Wang, H. Wu, and N.-F. Tzeng, "RFID Support for Accurate 3-Dimensional Localization," *IEEE Transactions on Computers*, vol. 62, no. 7, pp. 1447-1459, July 2013.
40. X.-Y. Xiang, I. Jangjaimon, M. Madani, and N.-F. Tzeng, "A Reliable and Cost-Effective Sand Monitoring System on the Field Programmable Gate Array (FPGA)," *IEEE Transactions on Instrumentation and Measurement*, vol. 62, no. 7, pp. 1870-1881, July 2013.
39. Adam W. Lewis, N.-F. Tzeng, and S. Ghosh, "Run-Time Energy Consumption Estimation for Server Workloads Based on Chaotic Time-Series Approximation," *ACM Transactions on Architecture and Code Optimization (TACO)*, vol. 9, issue 3, article no. 15, September 2012.
38. F. Pong and N.-F. Tzeng, "Concise Lookup Tables for IPv4 and IPv6 Longest Prefix Matching in Scalable Routers," *IEEE/ACM Transactions on Networking*, vol. 20, no. 3, pp. 729-741, June 2012.
37. F. Pong and N.-F. Tzeng, "HaRP: Rapid Packet Classification via Hashing Round-down Prefixes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 22, pp. 1105-1119, July 2011.
36. Paul J. Darby III and N.-F. Tzeng, "Decentralized QoS-Aware Checkpointing Arrangement in Mobile Grid Computing," *IEEE Transactions on Mobile Computing*, vol. 9, pp. 1173-1186, August 2010.
35. F. Pong and N.-F. Tzeng, "SUSE: Superior Storage-Efficiency for Routing Tables through Prefix Transformation and Aggregation," *IEEE/ACM Transactions on Networking*, vol. 18, no. 1, pp. 81-94, February 2010.
34. Y. Wang, H. Wu, F. Lin, and N.-F. Tzeng, "Cross-Layer Protocol Design and Optimization for Delay/Fault-Tolerant Mobile Sensor Networks (DFT-MSN's)," *IEEE Journal on Selected Areas in Communications*, vol. 26, no. 5, pp. 809-819, June 2008.
33. H. Chen, H. Wu, S. Kumar, and N.-F. Tzeng, "Minimum-Cost Data Delivery in Heterogeneous Wireless Networks," *IEEE Transactions on Vehicular Technology*, vol. 56, no. 6, pp. 3511-3523, November 2007.
32. N.-F. Tzeng, "Routing Table Partitioning for Speedy Packet Lookups in Scalable Routers," *IEEE Transactions on Parallel and Distributed Systems*, vol. 17, pp. 481-494, May 2006.

31. Y. Li, H. Wu, N.-F. Tzeng, D. Perkins, and M. Bayoumi, "MAC-SCC: Medium Access Control with a Separate Control Channel for Multihop Wireless Networks," *IEEE Transactions on Wireless Communications*, vol. 5, pp. 1805-1817, July 2006.
30. H. Wu, C. Wang, and N.-F. Tzeng, "Novel Self-Configurable Positioning Technique for Multihop Wireless Networks," *IEEE/ACM Transactions on Networking*, vol. 13, no. 3, pp. 609-621, June 2005.
29. H. Wu, A. Utgikar, and N.-F. Tzeng, "SYN-MAC: A Distributed MAC Protocol for Synchronized Wireless Networks," *Mobile Networks & Applications: The Journal of Special Issues on Mobility of Systems, Users, Data and Computing*, Springer, vol. 10, no. 5, pp. 627-637, October 2005.
28. N.-F. Tzeng, "Multistage-Based Switching Fabrics for Scalable Routers," *IEEE Transactions on Parallel and Distributed Systems*, vol. 15, pp. 304-318, April 2004.
27. S. S. Fu, N.-F. Tzeng and J.-Y. Chung, "Empirical Evaluation of Mutual Exclusion Algorithms for Distributed Systems," *Journal of Parallel and Distributed Computing*, vol. 60, pp. 785-806, July 2000.
26. N.-F. Tzeng and H.-L. Chen, "Fast Compaction in Hypercubes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 9, pp. 50-56, Jan. 1998.
25. H.-L. Chen and N.-F. Tzeng, "A Boolean Expression-Based Approach for Maximum Incomplete Subcube Identification in Faulty Hypercubes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 8, pp. 1171-1183, Nov. 1997.
24. H.-L. Chen and N.-F. Tzeng, "Subcube Determination in Faulty Hypercubes," *IEEE Transactions on Computers*, vol. 46, pp. 871-879, Aug. 1997.
23. S. S. Fu and N.-F. Tzeng, "A Circular List-Based Mutual Exclusion Scheme for Large Shared-Memory Multiprocessors," *IEEE Transactions on Parallel and Distributed Systems*, vol. 8, pp. 628-639, June 1997.
22. H.-L. Chen and N.-F. Tzeng, "On-Line Task Migration in Hypercubes through Double Disjoint Paths," *IEEE Transactions on Computers*, vol. 46, pp. 379-384, Mar. 1997.
21. N.-F. Tzeng and G. Lin, "Efficient Determination of Maximum Incomplete Subcubes in Hypercubes with Faults," *IEEE Transactions on Computers*, vol. 45, pp. 1303-1308, Nov. 1996.
20. N.-F. Tzeng and H. Kumar, "Traffic Analysis and Simulation Performance of Incomplete Hypercubes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 7, pp. 740-754, July 1996.
19. N.-F. Tzeng and G.-L. Feng, "Resource Allocation in Cube Network Systems Based on the Covering Radius," *IEEE Transactions on Parallel and Distributed Systems*, vol. 7, pp. 328-342, Apr. 1996.
18. G. Lin and N.-F. Tzeng, "Effective Utilization of Hypercubes in the Presence of Faults," *Journal of Parallel and Distributed Computing*, vol. 32, pp. 223-231, Feb. 1996.
17. N.-F. Tzeng and H.-L. Chen, "Structural and Tree Embedding Aspects of Incomplete Hypercubes," *IEEE Transactions on Computers*, vol. 43, pp. 1434-1439, Dec. 1994.
16. N.-F. Tzeng, "Reliable Butterfly Distributed-Memory Multiprocessors," *IEEE Transactions on Computers*, vol. 43, pp. 1004-1013, Sept. 1994.

15. N.-F. Tzeng and P.-J. Chuang, "A Pairwise Substitutional Fault Tolerance Technique for the Cube-Connected Cycles Architecture," *IEEE Transactions on Parallel and Distributed Systems*, vol. 5, pp. 433-439, Apr. 1994.
14. P.-J. Chuang and N.-F. Tzeng, "Allocating Precise Submeshes in Mesh Connected Systems," *IEEE Transactions on Parallel and Distributed Systems*, vol. 5, pp. 211-217, Feb. 1994.
13. H.-L. Chen and N.-F. Tzeng, "Efficient Resource Placement in Hypercubes Using Multiple-Adjacency Codes," *IEEE Transactions on Computers*, vol. 43, pp. 23-33, Jan. 1994.
12. N.-F. Tzeng, P.-J. Chuang, and C.-H. Wu, "Creating Disjoint Paths in Gamma Interconnection Networks," *IEEE Transactions on Computers*, vol. 42, pp. 1247-1252, Oct. 1993.
11. N.-F. Tzeng, "A Reliable Cube-Connected Cycles Structure," *Journal of Parallel and Distributed Computing*, vol. 19, pp. 64-71, Sept. 1993.
10. N.-F. Tzeng, "Reconfiguration and Analysis of a Fault-Tolerant Circular Butterfly Parallel System," *IEEE Transactions on Parallel and Distributed Systems*, vol. 4, pp. 855-863, Aug. 1993.
9. N.-F. Tzeng, "A Cube-Connected Cycles Architecture with High Reliability and Improved Performance," *IEEE Transactions on Computers*, vol. 42, pp. 246-253, Feb. 1993.
8. N.-F. Tzeng, "A Cost-Effective Combining Structure for Large-Scale Shared-Memory Multiprocessors," *IEEE Transactions on Computers*, vol. 41, pp. 1420-1429, Nov. 1992.
7. P.-J. Chuang and N.-F. Tzeng, "A Fast Recognition-Complete Processor Allocation Strategy for Hypercube Computers," *IEEE Transactions on Computers*, vol. 41, pp. 467-479, Apr. 1992.
6. N.-F. Tzeng and H.-L. Chen, "An Effective Approach to the Enhancement of Incomplete Hypercubes," *Journal of Parallel and Distributed Computing*, vol. 14, pp. 163-174, Feb. 1992.
5. S. K. Mishra, V. V. Raghavan, and N.-F. Tzeng, "Efficient Algorithms for Selection of Recovery Points in Tree Task Models," *IEEE Transactions on Software Engineering*, vol. 17, pp. 731-734, July 1991.
4. N.-F. Tzeng, "Alleviating the Impact of Tree Saturation on Multistage Interconnection Network Performance," *Journal of Parallel and Distributed Computing*, special issue on shared-memory multiprocessors, pp. 107-117, June 1991.
3. N.-F. Tzeng and S. Wei, "Enhanced Hypercubes," *IEEE Transactions on Computers*, vol. 40, pp. 284-294, Mar. 1991.
2. N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, "Realizing Fault-Tolerant Interconnection Networks Via Chaining," *IEEE Transactions on Computers*, special issue on fault-tolerant computing, vol. 37, pp. 458-462, Apr. 1988.
1. P.-C. Yew, N.-F. Tzeng, and D. H. Lawrie, "Distributing Hot-Spot Addressing in Large-Scale Multiprocessors," *IEEE Transactions on Computers*, special issue on parallel and distributed computing, vol. C-36, pp. 388-395, Apr. 1987.

Book Articles

- N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, “Realizing Fault-Tolerant Interconnection Networks Via Chaining,” *Advances in Distributed System Reliability*, Ed. by S. Rai and D. P. Agrawal, IEEE Computer Society Press, 1990, pp. 67-71.

Refereed Papers in Top Conference Proceedings

The following publications appeared in top quality conference proceedings after rigorously peer-reviewed. They typically had acceptance rates of no more than 30%.

69. S. Panta, R. M. Morvant, X. Yuan, L. Chen, and N.-F. Tzeng, “M3R: Localized Rainfall Nowcasting with Meteorology-Informed MultiModal Attention,” accepted in March 2026 for presentation at 2026 IEEE International Conference on Multimedia and Expo (ICME), July 2026.
68. Md S. Islam, I. Chapman, N. Ashafuddula, X. Yuan, L. Chen, N.-F. Tzeng, and K. Nahrstedt, “FedACT: A Resource Heterogeneity-Aware Device Scheduling Framework for Concurrent Training of Multiple Federated Learning Jobs,” *Proceedings of 40th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2026)*, May 2026.
67. Md S. Islam, S. Panta, X. Yuan, F. Xu, L. Chen, and N.-F. Tzeng, “SEAFL: Enhancing Efficiency in Semi-Asynchronous Federated Learning through Adaptive Aggregation and Selective Training,” *Proceedings of 39th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2025)*, June 2025, pp. 509 – 519.
66. J. Lou, X. Yuan, R. Zhang, X. L. Yuan, N. Gong, and N.-F. Tzeng, “GRID: Protecting Training Graph from Link Stealing Attacks on GNN Models,” accepted for presentation at 46th IEEE Symposium on Security and Privacy (IEEE S&P 2025), May 2025. (Acceptance Rate of 14.3%)
65. F. Lin, J. Lou, X. Yuan, and N.-F. Tzeng, “Towards Robust Vision Transformer via Masked Adaptive Ensemble,” *Proceedings of 33rd ACM International Conference on Information and Knowledge Management (CIKM 2024)*, pp. 1389-1399, October 2024.
64. M. S. Islam, S. Javaherian, F. Xu, X. Yuan, L. Chen, and N.-F. Tzeng, “FedClust: Tackling Data Heterogeneity in Federated Learning through Weight-Driven Client Clustering,” *Proceedings of 53rd International Conference on Parallel Processing (ICPP 2024)*, August 2024.
63. F. Lin, K. Guillot, S. Crawford, Y. Zhang, X. Yuan, and N.-F. Tzeng, “An Open and Large-Scale Dataset for Multi-Modal Climate Change-aware Crop Yield Predictions,” *Proceedings of 2024 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (2024 KDD)*, pp. 5375-5386, August 2024.
62. A. Sultana, L. Chen, X. Yuan, and N.-F. Tzeng, “Hadar: Heterogeneity-Aware Optimization-Based Online Scheduling for Deep Learning Clusters,” *Proceedings of 38th IEEE International Parallel & Distributed Processing Symposium (IPDPS)*, May 2024.
61. F. Lin, S. Crawford, K. Guillot, Y. Zhang, Y. Chen, X. Yuan, L. Chen, S. Williams, R. Minvielle, X. Xiao, D. Gholson, N. Ashwell, T. Setiyono, B. Tubana, L. Peng, M. Bayoumi, and N.-F. Tzeng, “MMST-ViT: Climate Change-aware Crop Yield Prediction via Multi-Modal Spatial-Temporal Vision Transformer,” *Proceedings of International Conference on Computer Vision (ICCV 2023)*, pp. 5774-5784, October 2023. (Acceptance Rate < 17%)
60. F. Lin, Y. Zhang, L. Chen, L. Peng, P. Sigdel, X. Yuan, and N.-F. Tzeng, “Comprehensive Transformer-based Model Architecture for Real-World Storm Prediction,” *Proceedings of*

European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2023), pp. 54-71, September 2023.

59. J. Lou, X. Yuan, M. Pan, H. Wang, and N.-F. Tzeng, "Data Privacy Examination against Semi-Supervised Learning," *Proceedings of 18th ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2023)*, July 2023.
58. F. Lin, X. Yuan, L. Peng, and N.-F. Tzeng, "Cascade Variational Auto-Encoder for Hierarchical Disentanglement," *Proceedings of 31st ACM International Conference on Information and Knowledge Management (CIKM)*, October 2022, pp. 1-10.
57. Y. Zhang, X. Yuan, and N.-F. Tzeng, "Platform-Oblivious Anti-Spam Gateway," *Proceedings of 37th Annual Computer Security Applications Conference (ACSAC)*, December 2021, pp. 1064-1077. (Acceptance rate of 24.5%)
56. Y. Zhang, J. Lou, L. Chen, J. Li, N.-F. Tzeng, and X. Yuan, "Reverse Attack: Black-box Attacks on Collaborative Recommendation," *Proceedings of ACM Conference on Computer and Communications Security (CCS 2021)*, November 2021, pp. 51-68. (Acceptance rate of 22%)
55. Y. Zhang, X. Yuan, S. Kimball, E. Rappin, L. Chen, P. Darby, T. Johnsten, L. Peng, B. Pitre, D. Bourrie, and N.-F. Tzeng, "Precise Weather Parameter Predictions for Target Regions via Neural Networks," *Proceedings of 2021 European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)*, September 2021. (Acceptance rate of 29%)
54. Y. He, F. Lin, X. Yuan, and N.-F. Tzeng, "Interpretable Minority Synthesis for Imbalanced Classification," accepted for presentation at *30th international Joint Conference on Artificial Intelligence (IJCAI-21)*, August 2021. (Acceptance rate of 13.9%)
53. Y. Zhang, J. Lou, L. Chen, X. Yuan, J. Li, T. Johnsten and N.-F. Tzeng, "Towards Poisoning the Neural Collaborative Filtering-Based Recommender Systems," *Proceedings of 25th European Symposium on Research in Computer Security (ESORICS 2020)*, September 2020. (Acceptance rate of 19.7%)
52. E. Beyazit, D. Tuncel, X. Yuan, N.-F. Tzeng, and X. Wu, "Learning Interpretable Representations with Informative Entanglements," *Proceedings of 29th International Joint Conference on Artificial Intelligence – Pacific Rim International Conference on Artificial Intelligence (IJCAI-PRICAI 2020)*, July 2020, pp. 1970-1976. (Acceptance rate of 12.6%)
51. J. Liu, Y. Zhang, J. Lian, X. Yuan, and N.-F. Tzeng, "Instant AoI Optimization in IoT Networks with Packet Combination," *Proceedings of 17th IEEE International Conference on Sensing, Communication and Networking (SECON 2020)*, June 2020.
50. E. Beyazit, Y. He, N.-F. Tzeng, and X. Wu, "Online Learning to Accelerate Neural Network Inference with Traveling Classifiers," *Proceedings of 24th European Conference on Artificial Intelligence (ECAI 2020)*, August/September 2020. (Acceptance rate of 26.8% for full papers)
49. J. Liu, X. Yuan, S. Kompella, and N.-F. Tzeng, "AoI and Throughput Tradeoffs in Routing-aware Multi-hop Wireless Networks," *Proceedings of 39th IEEE International Conference on Computer Communications (INFOCOM 2020)*, July 2020. (Acceptance rate of 19.8%)
48. Y. Zhang, H. Zhang, X. Yuan, and N.-F. Tzeng, "TweetScore: Scoring Tweets via Social Attribute Relationships for Twitter Spammer Detection," *Proceedings of 14th ACM Asia Conference on*

Information, Computer and Communications Security (ASIACCS 2019), July 2019, pp. 379-390. (Acceptance rate of 17.1% for full papers)

47. Y. Zhang, H. Zhang, X. Yuan, and N.-F. Tzeng, "Pseudo-honeypot: Toward Efficient and Scalable Spam Sniffer," *Proceedings of 49th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2019)*, June 2019, pp. 435-446. (Acceptance rate of 21.4%)
46. X.-Y. Xiang, S. Ghose, O. Mutlu, L. Peng, W. Shi, and N.-F. Tzeng, "Carpool: A Bufferless NoC with Adaptive Multicast and Hotspot Alleviation," *Proceedings of ACM Int'l Conference on Supercomputing (ICS '17)*, June 2017, Article No. 19. (Acceptance rate of 15.8%)
45. X.-Y. Xiang, S. Ghose, O. Mutlu, and N.-F. Tzeng, "A Model for Application Slowdown Estimation in On-Chip Networks and Its Use for Improving System Fairness and Performance," *Proceedings of 34th IEEE Int'l Conference on Computer Design (ICCD 2016)*, October 2016, pp. 456-463.
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29. N.-F. Tzeng and A. Kongmunvattana, "Distributed Shared Memory Systems with Improved Barrier Synchronization and Data Transfer," *Proc. 11th ACM Int'l Conf. Supercomputing*, July 1997, pp. 148-155.
28. S. S. Fu and N.-F. Tzeng, "Aggressive Release Consistency for Software Distributed Shared Memory," *Proc. 17th Int'l Conf. Distributed Computing Systems*, May 1997, pp. 288-295.
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26. N.-F. Tzeng and G. Lin, "Maximum Reconfiguration of 2-D Mesh Systems with Faults," *Proc. 1996 Int'l Conf. Parallel Processing*, Aug. 1996, vol. I, pp. 77-84.
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24. G. Lin and N.-F. Tzeng, "Reconfiguration and Experiments in a Faulty Hypercube," *Proc. 9th Int'l Parallel Processing Symposium*, Apr. 1995, pp. 592-598.
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20. N.-F. Tzeng and G.-L. Feng, "On Resource Allocation in Binary n -Cube Network Systems," *Proc. 22nd Int'l Conf. Parallel Processing*, Aug. 1993, vol. II, pp. 209-212.
19. H.-L. Chen and N.-F. Tzeng, "Quick Determination of Subcubes in a Faulty Hypercube," *Proc. 21st Int'l Conf. Parallel Processing*, Aug. 1992, vol. III, pp. 338-345.
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13. P.-J. Chuang and N.-F. Tzeng, "An Efficient Submesh Allocation Strategy for Mesh Computer Systems," *Proc. 11th IEEE Int'l Conf. Distributed Computing Systems*, May 1991, pp. 256-263.
12. C.-H. Chen and N.-F. Tzeng, "An Area-Efficient Reconfigurable Binary Tree Architecture," *Proc. 1990 IEEE Int'l Conf. Computer Design – VLSI in Computers and Processors*, Sept. 1990, pp. 376-379.
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8. P.-J. Chuang and N.-F. Tzeng, "Dynamic Processor Allocation in Hypercube Computers," *Proc. 17th IEEE/ACM Int'l Symp. Computer Architecture*, May 1990, pp. 40-49.
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6. N.-F. Tzeng, "Design of a Novel Combining Structure for Shared-Memory Multiprocessors," *Proc. 1989 Int'l Conf. Parallel Processing*, Aug. 1989, pp. I 1-8.
5. H.-L. Chen and N.-F. Tzeng, "Enhanced Incomplete Hypercubes," *Proc. 1989 Int'l Conf. Parallel Processing*, Aug. 1989, vol. I, pp. 270-277.
4. P.-C. Yew, N.-F. Tzeng, and D. H. Lawrie, "Distributing Hot-Spot Addressing in Large-Scale Multiprocessors," *Proc. 1986 IEEE Int'l Conf. Parallel Processing*, Aug. 1986, pp. 51-58.
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Refereed Papers in Other Conference Proceedings

The following papers were reviewed and presented in conferences, whose accepted rates typically ranged from 30% to 50%.

58. S. Panta, X. Yuan, L. Chen, and N.-F. Tzeng, "Revisiting the Seasonal Trend Decomposition for Enhanced Time Series Forecasting," accepted for presentation at 51th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026), May 2026.
57. C. Jiang, Z. Zhao, P. Liang, M. Shi, J. Han, N.-F. Tzeng, G. Xiao, D. Chen, and H. Zheng, "Exploring Multi-Scale Local and Global Features in Whole Slide Images Using State Space Models," accepted for presentation at IEEE 23rd International Symposium on Biomedical Imaging (ISBI 2026), April 2026.
56. R. Pradhananga, S. William, S. Aygun, L. Chen, Y. Tu, W. Crow, S. Aakur, and N.-F. Tzeng, "Digital Twin-Aided Municipal Traffic Control," *Proceedings of 2025 SUMO User Conference*, May 2025.
55. F. Qi, X. Fu, X. Yuan, N.-F. Tzeng, and L. Peng, "A New Routing Strategy to Improve Success Rates of Quantum Computers," *Proceedings of 34th ACM Great Lakes Symposium on VLSI (GLSVLSI 2024)*, June 2024.
54. C. Vasquez, T. LeCompte, X. Yuan, N.-F. Tzeng, and L. Peng, "Soft Error Resilience Analysis of LSTM Networks," *Proceedings of 34th ACM Great Lakes Symposium on VLSI (GLSVLSI 2024)*, June 2024.
53. Md S. Islam, S. Javaherian, F. Xu, X. Yuan, L. Chen, and N.-F. Tzeng, "FedClust: Optimizing Federated Learning on Non-IID Data through Weight-Driven Client Clustering," *Proceedings of 38th IEEE International Parallel and Distributed Processing Symposium (IPDPS 2024)*, Poster Presentation, May 2024.
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45. A. Lewis, J. Simon, and N.-F. Tzeng, "Chaotic Attractor Prediction of Run-time Energy Consumption Based on Workload in Server Systems," *Proceedings of USENIX 2010 Workshop on Power Aware Computing and Systems (HotPower '10)*, October 2010.
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31. F. Pong and N.-F. Tzeng, "Storage-Efficient Architecture for Routing Tables via Prefix Transformation," *Proceedings of 32nd IEEE Conference on Local Computer Networks (LCN 2007)*, October 2007, pp. 55-62.
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27. Paul J. Darby III and N.-F. Tzeng, "Peer-to-Peer Checkpointing Arrangement for Mobile Grid Computing Systems," *Proc. of IEEE International Conference on High-Performance Distributed Computing (HPDC-2007)*, June 2007.
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25. Y. Wang, H. Wu, R. Nelavelli, and N.-F. Tzeng, "Balance-Based Energy-Efficient Communication Protocols for Wireless Sensor Networks," *Proc. Workshop on Wireless Ad-hoc and Sensor Networks* (held in conjunction with 26th IEEE Int'l Conference on Distributed Computing Systems, *ICDCS '06*), July 2006.
24. G. Wang and N.-F. Tzeng, "TCAM-Based Forwarding Engine with Minimum Independent Prefix Set (MIPS) for Fast Updating," *Proc. 2006 IEEE Int'l Conf. on Communications (ICC '06)*, June 2006, pp. 103-109.
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21. N.-F. Tzeng, "STRESS: Efficient Multicast Shared Trees via Restricted Search," *Proc. 2005 IEEE Int'l Conf. on Communications (ICC '05)*, May 2005, pp. 965-971.
20. N.-F. Tzeng, "SPAL: A Speedy Packet Lookup Technique for High-Performance Routers," *Proc. 33rd Int'l Conf. on Parallel Processing*, Aug. 2004, pp. 284-291.

19. M. Mandviwalla and N.-F. Tzeng, "DRA: A Dependable Architecture for High-Performance Routers," *Proc. Int'l Workshop on Network Design and Architecture (IWNA 2004, held in conjunction with 33rd Int'l Conf. on Parallel Processing)*, Aug. 2004, pp. 265-272.
18. N.-F. Tzeng, "Hardware-Assisted Design for Fast Packet Forwarding in Parallel Routers," *Proc. 32nd Int'l Conf. on Parallel Processing*, Oct. 2003, pp. 11-18.
17. Y. Li, H. Wu, D. Perkins, N.-F. Tzeng, and M. Bayoumi, "MAC-SCC: Medium Access Control with a Separate Control Channel for Multihop Wireless Networks," *Proc. Workshop on Mobile and Wireless Networks (MWN '03)*, held in conjunction with 23rd IEEE Int'l Conf. on Distributed Computing Systems, May 2003, pp. 764-769.
16. N.-F. Tzeng and P. Alla, "Guided Shared Trees for Efficient Multicast in Large Networks," *Proc. 2003 IEEE Int'l Conf. on Communications (ICC '03)*, May 2003, pp. 87-92.
15. N.-F. Tzeng and R. Batchu, "Design and Evaluation of Scalable Switching Fabrics for High-Performance Routers," *Proc. 2002 Int'l Conf. on Parallel Processing*, Aug. 2002, pp. 167-174.
14. N.-F. Tzeng and M. Mandviwalla, "Performance Evaluation of Router Switching Fabrics," *Proc. 9th International Conference on Parallel and Distributed Systems*, Dec. 2002, pp. 542-547.
13. A. Ganjoo and N.-F. Tzeng, "Influence of High-Level Program Structures on Branch Prediction Accuracy," *Proc. Euromicro Conf. on Digital System Design*, Sept. 2000, vol. 1, pp. 316-322.
12. N.-F. Tzeng, K. Ponnuru, and K. Vibhatavanij, "A Cost-Effective Design for ATM Switching Fabrics," *Proc. 1999 IEEE Int'l Conference on Communications (ICC '99)*, June 1999, pp. S37.4.1-5.
11. S. Fu and N.-F. Tzeng, "Lock Improvement Scheme for Release Consistency in Distributed Shared Memory Systems," *Proc. 6th Symp. Frontiers of Massively Parallel Computation*, Oct. 1996, pp. 255-262.
10. Y. Liao, M. Lu, N.-F. Tzeng, "The Parilindrome Network for Fault-Tolerant Interconnection," *Proc. 8th IEEE Symp. Parallel and Distributed Processing*, Oct. 1996, pp. 556-561.
9. C.-H. Wu, F. F. Young, and N.-F. Tzeng, "Shortening Turnaround Time for Two-Dimensional Plasma Processing Simulations via Vectorization and Parallelization Techniques," *Proc. 7th SIAM Conf. Parallel Processing for Scientific Computing*, Feb. 1995, pp. 237-238.
8. H.-L. Chen and N.-F. Tzeng, "Task Migration in Hypercubes Using All Disjoint Paths," *Proc. 7th Int'l Conf. Parallel and Distributed Computing Systems*, Oct. 1994, pp. 358-363.
7. N.-F. Tzeng and G. Lin, "Identifying Maximal Incomplete Subcubes in Faulty Hypercubes," *Proc. 7th Int'l Conf. Parallel and Distributed Computing Systems*, Oct. 1994, pp. 186-193.
6. N.-F. Tzeng and H. Kumar, "Analysis of Link Traffic in Incomplete Hypercubes," *Proc. 5th IEEE Symp. Parallel and Distributed Processing*, Dec. 1993, pp. 312-319.
5. N.-F. Tzeng, "An Alternative Combining Architecture for Large-Scale Shared-Memory Multiprocessors," *Proc. 1991 IEEE Int'l Symp. Circuits and Systems*, June 1991, pp. 1049-1052.
4. N.-F. Tzeng, "A Reconfigurable Cube-Connected Cycles Architecture for Wafer Scale Integration," *Proc. IEEE Int'l Conf. Wafer Scale Integration*, Jan. 1991, pp. 33-39.
3. N.-F. Tzeng and C.-H. Chen, "Yield Evaluation of WSI Parallel Systems," *Proc. IEEE Int'l Conf. Wafer Scale Integration*, Jan. 1991, pp. 156-162.

2. N.-F. Tzeng, "An Effective Fault-Tolerant Technique for Circular Butterfly Parallel Systems," *Proc. 2nd IEEE Symp. Parallel & Distributed Processing*, Dec. 1990, pp. 223-226.
1. N.-F. Tzeng and S. Bhattacharya, "Binary Tree Layouts with Adequate I/O Support," *Proc. 1990 IEEE Int'l Symp. Circuits and Systems*, May 1990.

PROFESSIONAL HONORS AND ACTIVITIES

- Recipient, University Foundation Distinguished Professor Award, 1997.
- Recipient, the Outstanding Paper Award of 10th IEEE International Conference on Distributed Computing Systems, May 1990.
- Fellow, IEEE, 2010.
- Distinguished Member, ACM (Association for Computing Machinery), 2022.
- Chair, Technical Committee on Distributed Processing, IEEE Computer Society (Elected, 1999 – 2002).
- Editor, IEEE Transactions on Parallel and Distributed Systems (1998 – 2001).
- Editor, IEEE Transactions on Computers (1994 – 1998).
- Editor, Journal of Information Science and Engineering (JISE, published by Academia Sinica, Taiwan), 2011 – 2014.
- Co-Guest Editor, special issue of the Journal of Parallel and Distributed Computing on distributed shared memory systems, September 1995.
- Member, IEEE Fellows Evaluation Committee, IEEE Computer Society, 2010, 2011, 2013, 2014, 2015, 2016, 2018, 2019, 2020, 2023, 2026.
- Chair, Award Selection Subcommittee for IEEE Computer Society's Edward J. McCluskey Technical Achievement Award, 2024, 2025.
- Member, Award Selection Subcommittee for IEEE Computer Society's Edward J. McCluskey Technical Achievement Award, 2022, 2023
- Member, 2023 Editor-in-Chief Selection Committee for IEEE Transactions on Computers
- Member, 2024 Editor-in-Chief Selection Committee for IEEE Transactions on Sustainable Computing
- Panelist for proposal evaluation, NSF Division of Computing and Communication Foundations, 2008, 2009, 2010, 2011, 2012, 2013, February 2014, June 2014, 2015, 2017.
- Panelist for proposal evaluation, NSF Division of Computer-Communications Research, 1999, 2000, 2002, and 2003.
- Panelist for proposal evaluation, NSF Information Technology Research (ITR) Program, 2000.
- Panelist for proposal evaluation, NSF POWRE Program, Division of Experimental and Integrative Activities, 1999.
- Distinguished Visitor, IEEE Computer Society (1994 – 1997).
- Newsletter Editor, Technical Committee on Distributed Processing, IEEE Computer Society (1995 - 1997).

- Technical Program Vice-Chair (for Cyber Physical Systems), 45th International Conference on Parallel Processing (ICPP 2016), August 2016.
- Technical Program Chair, 10th International Conference on Parallel and Distributed Systems (ICPADS 2004), July 2004.
- Award Co-Chair, 2006 IEEE Symposium on Applications and the Internet (SAINT-2006), January 2006.
- Award Co-Chair, 11th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2005), July 2005.
- Chair, Outstanding Achievement Award Committee, IEEE Technical Committee on Distributed Processing (TCDP), 2007.
- Workshop Co-Chair, 24th IEEE International Conference on Distributed Computing Systems (ICDCS 2004), March 2004.
- Member, technical program committee of 15th IEEE International Conference on Networking, Architecture, and Storage (NAS 2021), October 2021.
- Member, Awards Committee, 30th IEEE International Conference on Distributed Computing Systems (ICDCS 2010), June 2010.
- Member, Steering Committee, 11th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2005), July 2005.
- Member, Outstanding Achievement Award Committee, IEEE Technical Committee on Distributed Processing (TCDP), 2006.
- Member, Steering Committee, IEEE International Conference on Mobile Ad-hoc and Sensor Systems, 2004 - 2008.
- Member, Computer Entrepreneur Award Committee, the IEEE Computer Society, 2003, 2005.
- International Liaison Co-Chair, 2003 Symposium on Applications and the Internet (SAINT-2003), January 2003.
- Workshops Chair, 2002 Symposium on Applications and the Internet (SAINT-2002), January - February 2002.
- Tutorials Chair, 2002 IEEE International Conference on Parallel and Distributed Systems (ICPAD-2002), December 2002.
- Keynote Speaker, 2005 Workshop on Wireless, Ad Hoc, and Sensor Networks (WASN 2005), August 2005.
- Member, technical program committee of 28th International Conference on Distributed Computing Systems (ICDCS 2008), June 2008.
- Member, technical program committee of 14th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC 2008), Dec. 2008.
- Member, technical program committee of 21st IEEE Symposium on Reliable Distributed Systems (SRDS), October 2002.
- Member, technical program committee of 20th IEEE Symposium on Reliable Distributed Systems (SRDS), October 2001.
- Member, technical program committee of 2001 International Conference on Computer Networks and Mobile Computing, October 2001.

- Member, technical program committee of 5th International Conference on Computer Science and Informatics (CS&I), February 2000.
- Member, technical program committee of International Conference on High Performance Computing (HiPC'99), December 1999.
- Member, technical program committee of International Conference on Parallel and Distributed Systems (ICPADS'98), December 1998.
- Member, technical program committee of International Conference on Computer Design, October 1997.
- Member, technical program committee of International Conference on Computer Design – VLSI in Computers and Processors, October 1996.
- Member, technical program committee of International Conference on High Performance Computing, December 1995.
- Member, technical program committee of 9th ACM International Conference on Supercomputing, July 1995.
- Member, technical program committee of 1st International Symposium on High Performance Computer Architecture, January 1995.
- Member, technical program committee of 7th International Conference on Parallel and Distributed Computing Systems, October 1994.
- Member, technical program committee of 11th International Conference on Distributed Computing Systems, May 1991.
- Organizer, panel on "Distributed Shared Memory Systems" at 1996 IEEE International Conference on Computer Design: VLSI in Computers & Processors (ICCD '96), October 1996.
- Session Chair for: 19th International Conference on Parallel Processing, August 1990; 20th International Conference on Parallel Processing, August 1991; 22nd International Conference on Parallel Processing, August 1993; 5th IEEE Symposium on Parallel and Distributed Processing, December, 1993; 23rd International Conference on Parallel Processing, August 1994; 28th International Conference on Parallel Processing, September 1999; 20th IEEE Symposium on Reliable Distributed Systems (SRDS), October 2001; 2006 Symposium on Applications and the Internet (SAINT 2006), January 2006; 6th IEEE International Symposium on Network Computing and Applications, July 2007.
- External Examiner, Nanyang Technological University, Singapore, 2001, 2002.
- Reviewer, book manuscript under consideration by publisher, John Wiley & Sons, Inc., 2002.
- Referee, National Science Foundation (NSF), IEEE Transactions on Computers, IEEE Transactions on Parallel and Distributed Systems, IEEE/ACM Transactions on Networking, IEEE Transactions on Software Engineering, Journal of Parallel and Distributed Computing, IEEE Transactions on VLSI Systems, IEEE Transactions on Reliability, IEEE Transactions on Circuits and Systems, Part II, IEEE Communications Letters, Distributed Computing, Parallel Computing, Journal of Supercomputing, Journal of Information Science and Engineering.
- Referee, IEEE Computer magazine, IEEE Parallel & Distributed Technology: Systems and Applications, ACM Computing Reviews, IEEE Communication Letters, Information Processing Letters, Parallel Processing Letters, Computing Reviews.
- Referee, many international conferences and symposiums.

● **Invited Lectures to:**

- ◇ Department of Electrical Engineering, Auburn University, May 1991
- ◇ Department of Computer Science, University of Houston, March 1992
- ◇ Department of Computer and Information Science, National Chiao-Tung University, Taiwan, May 1992
- ◇ Department of Computer Science, National Tsing Hua University, Taiwan, May 1992
- ◇ Department of Computer Science, Texas A&M University, November 1993
- ◇ Department of Electrical and Computer Engineering, Iowa State University, March 1994
- ◇ School of Electrical and Computer Engineering, Georgia Institute of Technology, May 1994
- ◇ Department of Electrical and Computer Engineering, State University of New York at Buffalo, October 1995
- ◇ University of Arkansas at Little Rock, September 1996
- ◇ Department of Electrical Engineering, University of Texas at San Antonio, October 1996
- ◇ Department of Computer Science, National Tsing Hwa University, Taiwan, April 2000.
- ◇ Department of Computer Science and Engineering, University of Texas, Arlington, October 2017.